

```
In [10]: import pandas as pd
df = pd.read_csv("customer_orders.csv")
df
```

```
Out[10]:
```

	OrderID	CustomerName	Product	Quantity	PricePerUnit
0	101	John Doe	Laptop	1.0	1200.50
1	102	Jane Smith	Phone	2.0	699.99
2	103	NaN	TV	1.0	45000.00
3	104	Mike Brown	Tablet	1.0	20000.00
4	105	Alice Johnson	Speaker	3.0	1500.00
5	105	Alice Johnson	Speaker	3.0	1500.00
6	106	NaN	Smartwatch	-1.0	2999.00
7	107	David Clark	Headphones	2.0	5000.00
8	108	Eve Black	Monitor	1.0	NaN
9	109	Frank White	Keyboard	1.0	200.00
10	110	Emily Green	Mouse	NaN	50.00
11	111	Tom Lee	Smartwatch	2.0	2500.00
12	112	Nancy Moore	Earphones	1.0	299.99
13	113	John Snow	Laptop	3.0	75000.00
14	114	Emma Stone	Phone	1.0	999.99
15	115	Daniel Craig	Tablet	4.0	NaN
16	116	Sophia Turner	Speaker	2.0	1999.00
17	117	Michael Scott	NaN	1.0	3500.00
18	118	Jim Halpert	Headphones	2.0	1500.00
19	119	Pam Beesly	Keyboard	3.0	1200.00
20	120	Dwight Schrute	Mouse	2.0	399.99

```
In [12]: df.isnull().sum()
```

```
Out[12]: OrderID      0
CustomerName  0
Product      1
Quantity     1
PricePerUnit  2
dtype: int64
```

```
In [20]: df['CustomerName'] = df['CustomerName'].fillna('Unknown')
df
```

C:\Users\LENOVO\AppData\Local\Temp\ipykernel_18880\909436581.py:1: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.
Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df['CustomerName'] = df['CustomerName'].fillna('Unknown')
```

```
Out[20]:
```

	OrderID	CustomerName	Product	Quantity	PricePerUnit
0	101	John Doe	Laptop	1.0	1200.500000
1	102	Jane Smith	Phone	2.0	699.990000
2	103	Unknown	TV	1.0	45000.000000
3	104	Mike Brown	Tablet	1.0	20000.000000
4	105	Alice Johnson	Speaker	3.0	1500.000000
5	105	Alice Johnson	Speaker	3.0	1500.000000
6	106	Unknown	Smartwatch	-1.0	2999.000000
7	107	David Clark	Headphones	2.0	5000.000000
8	108	Eve Black	Monitor	1.0	9194.358889
9	109	Frank White	Keyboard	1.0	200.000000
11	111	Tom Lee	Smartwatch	2.0	2500.000000
12	112	Nancy Moore	Earphones	1.0	299.990000
13	113	John Snow	Laptop	3.0	75000.000000
14	114	Emma Stone	Phone	1.0	999.990000
15	115	Daniel Craig	Tablet	4.0	9194.358889
16	116	Sophia Turner	Speaker	2.0	1999.000000
17	117	Michael Scott	Unknown Product	1.0	3500.000000
18	118	Jim Halpert	Headphones	2.0	1500.000000
19	119	Pam Beesly	Keyboard	3.0	1200.000000
20	120	Dwight Schrute	Mouse	2.0	399.990000

```
In [21]: df['Product'] = df['Product'].fillna('Unknown Product')
df
```

C:\Users\LENOVO\AppData\Local\Temp\ipykernel_18880\3063104082.py:1: SettingWithCopyWarning:
 A value is trying to be set on a copy of a slice from a DataFrame.
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```
df['Product'] = df['Product'].fillna('Unknown Product')
```

Out[21]:

	OrderID	CustomerName	Product	Quantity	PricePerUnit
0	101	John Doe	Laptop	1.0	1200.500000
1	102	Jane Smith	Phone	2.0	699.990000
2	103	Unknown	TV	1.0	45000.000000
3	104	Mike Brown	Tablet	1.0	20000.000000
4	105	Alice Johnson	Speaker	3.0	1500.000000
5	105	Alice Johnson	Speaker	3.0	1500.000000
6	106	Unknown	Smartwatch	-1.0	2999.000000
7	107	David Clark	Headphones	2.0	5000.000000
8	108	Eve Black	Monitor	1.0	9194.358889
9	109	Frank White	Keyboard	1.0	200.000000
11	111	Tom Lee	Smartwatch	2.0	2500.000000
12	112	Nancy Moore	Earphones	1.0	299.990000
13	113	John Snow	Laptop	3.0	75000.000000
14	114	Emma Stone	Phone	1.0	999.990000
15	115	Daniel Craig	Tablet	4.0	9194.358889
16	116	Sophia Turner	Speaker	2.0	1999.000000
17	117	Michael Scott	Unknown Product	1.0	3500.000000
18	118	Jim Halpert	Headphones	2.0	1500.000000
19	119	Pam Beesly	Keyboard	3.0	1200.000000
20	120	Dwight Schrute	Mouse	2.0	399.990000

In [22]:

```
df = df.dropna(subset=['Quantity'])
df
```

Out[22]:

	OrderID	CustomerName	Product	Quantity	PricePerUnit
0	101	John Doe	Laptop	1.0	1200.500000
1	102	Jane Smith	Phone	2.0	699.990000
2	103	Unknown	TV	1.0	45000.000000
3	104	Mike Brown	Tablet	1.0	20000.000000
4	105	Alice Johnson	Speaker	3.0	1500.000000
5	105	Alice Johnson	Speaker	3.0	1500.000000
6	106	Unknown	Smartwatch	-1.0	2999.000000
7	107	David Clark	Headphones	2.0	5000.000000
8	108	Eve Black	Monitor	1.0	9194.358889
9	109	Frank White	Keyboard	1.0	200.000000
11	111	Tom Lee	Smartwatch	2.0	2500.000000
12	112	Nancy Moore	Earphones	1.0	299.990000
13	113	John Snow	Laptop	3.0	75000.000000
14	114	Emma Stone	Phone	1.0	999.990000
15	115	Daniel Craig	Tablet	4.0	9194.358889
16	116	Sophia Turner	Speaker	2.0	1999.000000
17	117	Michael Scott	Unknown Product	1.0	3500.000000
18	118	Jim Halpert	Headphones	2.0	1500.000000
19	119	Pam Beesly	Keyboard	3.0	1200.000000
20	120	Dwight Schrute	Mouse	2.0	399.990000

```
In [24]: mean_price = df['PricePerUnit'].mean()
df['PricePerUnit'] = df['PricePerUnit'].fillna(mean_price)

print("Mean PricePerUnit:", mean_price)
```

Mean PricePerUnit: 9194.358888888888

```
In [27]: invalid_values = df[df['Quantity'] < 0]
print("\nInvalid Quantity values:")
print(invalid_values['Quantity'])
df.loc[df['Quantity'] < 0, 'Quantity'] = (
    df.loc[df['Quantity'] < 0, 'Quantity'].abs()
)
```

Invalid Quantity values:

6 -1.0

Name: Quantity, dtype: float64

```
In [28]: duplicate_count = df.duplicated().sum()
print("\nNumber of duplicate rows:", duplicate_count)
df = df.drop_duplicates()
```

Number of duplicate rows: 1

```
In [30]: df['Quantity'] = df['Quantity'].astype(int)
df
```

C:\Users\LENOVO\AppData\Local\Temp\ipykernel_18880\3384934941.py:1: SettingWithCopyWarning:
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See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
df['Quantity'] = df['Quantity'].astype(int)

```
Out[30]:
```

	OrderID	CustomerName	Product	Quantity	PricePerUnit
0	101	John Doe	Laptop	1	1200.500000
1	102	Jane Smith	Phone	2	699.990000
2	103	Unknown	TV	1	45000.000000
3	104	Mike Brown	Tablet	1	20000.000000
4	105	Alice Johnson	Speaker	3	1500.000000
6	106	Unknown	Smartwatch	1	2999.000000
7	107	David Clark	Headphones	2	5000.000000
8	108	Eve Black	Monitor	1	9194.358889
9	109	Frank White	Keyboard	1	200.000000
11	111	Tom Lee	Smartwatch	2	2500.000000
12	112	Nancy Moore	Earphones	1	299.990000
13	113	John Snow	Laptop	3	75000.000000
14	114	Emma Stone	Phone	1	999.990000
15	115	Daniel Craig	Tablet	4	9194.358889
16	116	Sophia Turner	Speaker	2	1999.000000
17	117	Michael Scott	Unknown Product	1	3500.000000
18	118	Jim Halpert	Headphones	2	1500.000000
19	119	Pam Beesly	Keyboard	3	1200.000000
20	120	Dwight Schrute	Mouse	2	399.990000

```
In [32]: df['TotalPrice'] = df['Quantity'] * df['PricePerUnit']
```

```
df
```

```
C:\Users\LENOVO\AppData\Local\Temp\ipykernel_18880\1471182885.py:1: SettingWithCopyWarning:
```

```
A value is trying to be set on a copy of a slice from a DataFrame.
```

```
Try using .loc[row_indexer,col_indexer] = value instead
```

```
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
```

```
df['TotalPrice'] = df['Quantity'] * df['PricePerUnit']
```

```
Out[32]:
```

	OrderID	CustomerName	Product	Quantity	PricePerUnit	TotalPrice
0	101	John Doe	Laptop	1	1200.500000	1200.500000
1	102	Jane Smith	Phone	2	699.990000	1399.980000
2	103	Unknown	TV	1	45000.000000	45000.000000
3	104	Mike Brown	Tablet	1	20000.000000	20000.000000
4	105	Alice Johnson	Speaker	3	1500.000000	4500.000000
6	106	Unknown	Smartwatch	1	2999.000000	2999.000000
7	107	David Clark	Headphones	2	5000.000000	10000.000000
8	108	Eve Black	Monitor	1	9194.358889	9194.358889
9	109	Frank White	Keyboard	1	200.000000	200.000000
11	111	Tom Lee	Smartwatch	2	2500.000000	5000.000000
12	112	Nancy Moore	Earphones	1	299.990000	299.990000
13	113	John Snow	Laptop	3	75000.000000	225000.000000
14	114	Emma Stone	Phone	1	999.990000	999.990000
15	115	Daniel Craig	Tablet	4	9194.358889	36777.435556
16	116	Sophia Turner	Speaker	2	1999.000000	3998.000000
17	117	Michael Scott	Unknown Product	1	3500.000000	3500.000000
18	118	Jim Halpert	Headphones	2	1500.000000	3000.000000
19	119	Pam Beesly	Keyboard	3	1200.000000	3600.000000
20	120	Dwight Schrute	Mouse	2	399.990000	799.980000

```
In [ ]:
```